

The Living World and Classification

Science Unit Test One

Question 5: Adaptation Analysis

1. How can plants like Cactus and Acacia live in deserts with scarce water?

Cactus and Acacia plants withstand hot and dry desert environments through various structural modifications. They possess a thick cuticle on their leaf surfaces and stomata arranged in deep pits to minimize water loss by transpiration. Additionally, they utilize a specialized photosynthetic pathway (CAM) where stomata remain closed during the day to conserve moisture.

2. What adaptations do you see in the plants of snowy regions?

Plants in snowy regions have evolved structural and chemical adaptations to survive freezing temperatures, harsh winds, and heavy snow. Their conical shape and sloping branches allow snow to slide off easily, preventing damage, while thick bark helps them withstand extreme cold.

3. How is a frog adapted to an amphibious habitat? Why is it called an amphibian?

A frog is called an amphibian because it can live both on land and in water. It is adapted to this dual habitat through moist skin that allows for respiration in water and lungs for breathing on land. Furthermore, their webbed feet facilitate efficient swimming, and their powerful hind legs are specialized for leaping on land.

4. Why is the camel called the 'Ship of the Desert'?

The camel is known as the 'Ship of the Desert' due to its ability to traverse long distances across sandy terrain. Its padded feet prevent it from sinking into the sand, and it possesses the ability to store water and food (as fat in its hump), allowing it to survive for days in harsh, resource-scarce conditions.

Question 6: Comparison of Herbivores and Carnivores

Feature	Herbivores	Carnivores
Primary Diet	Plants (e.g., cows, deer)	Other animals (e.g., lions, wolves)
Dental Structure	Flat, broad teeth for grinding plant matter.	Sharp, pointed teeth for tearing flesh.
Digestive System	Long digestive tracts to process complex cellulose.	Shorter digestive tracts as meat is easier to digest.
Vision	Eyes on the sides for a wide field of view to spot predators.	Eyes in the front for depth perception during hunting.
Trophic Level	Primary consumers.	Secondary or tertiary consumers.
Social Behavior	Often live in herds for protection.	Often solitary or in small hunting packs/prides.

Question 7:

- **Extreme heat in deserts:** High temperatures occur due to intense sunlight and minimal cloud cover. Organisms adapt via thick cuticles, CAM photosynthesis, or nocturnal behavior to avoid peak heat.

- **Lush green grasslands:** These areas receive sufficient rainfall to support dense grass growth, which provides abundant food and shelter for a diverse range of herbivores.
- **Large insect populations:** Insects are highly adaptable due to their small size, rapid reproduction cycles, and ability to thrive on various food sources across almost every ecosystem.
- **Hiding as a defense:** Many organisms use camouflage, burrowing, or staying in dark, sheltered places to avoid detection by predators and ensure survival.
- **Long ears as an adaptation:** Large ears, seen in rabbits or fennec foxes, assist in thermoregulation by dissipating body heat and improve hearing sensitivity to detect threats.

Question 8: Use your brain power

1. **Why does water trickle off lotus leaves?**

Ans. Because of a waxy, water-repellent coating on the leaf surface.

2. **Why don't the leaves of these plants rot in water?**

Ans. They have a waxy, protective layer that prevents water absorption and decay.

3. **Why are their roots short and fibrous?**

Ans. Since they live in water, they absorb nutrients directly through their surface and do not need deep roots to search for water.

Question 9: Give scientific reason

The petiole and stem of aquatic plants have air spaces.

Ans. These air spaces provide buoyancy, helping the plant float, and facilitate the diffusion of gases like oxygen and carbon dioxide throughout the plant.

2. Trees and creepers in the jungle grow tall.

Ans. They grow tall to compete for and reach sunlight, which is limited in the dense jungle understory.

3. There are haustorial roots in Cuscuta.

Ans. Cuscuta is a parasitic plant that lacks chlorophyll; its haustorial roots penetrate the host plant's tissues to absorb water and nutrients.

4. Fungus cannot perform photosynthesis.

Ans. Fungi lack chlorophyll, the pigment necessary to capture light energy for photosynthesis.

5. The fishes have gills instead of nose.

Ans. Fishes live in water and gills are specialized organs that allow them to extract dissolved oxygen directly from water, which is more efficient than breathing air.

6. The prey is unaware about tiger approaching.

Ans. Tigers have evolved excellent camouflage (stripes) that blends with the forest background, and they move silently, allowing them to approach prey undetected.